

03/13/2026

Spectral algebraic

geometry I

Last time Comparison to derived ∞ -categories, graded homotopy groups of spectra, (faithful) flatness

Today Descent vs hyperdescent, quasicoherent sheaves, quotient stacks & equivariant sheaves

After spring break. Begin the proof of toric mirror symmetry

Recall

(1) we have lax symmetric monoidal functors

$$\mathrm{Mod}_{\mathbb{Z}}^{\heartsuit} \hookrightarrow D(\mathbb{Z}) \xrightarrow{H} Sp$$

For an ordinary ring A , this induces a symmetric monoidal equiv

$$\mathrm{Mod}_{\#A}(Sp) \xrightarrow{\sim} D(A)$$

(2) If A is an E_{∞} -ring, then $\pi_*(A) = \bigoplus_{d \in \mathbb{Z}} \pi_d(A)$ is naturally a graded commutative ring.

- We have the Koszul Sign rule if $a \in \pi_m(A)$ and $b \in \pi_n(A)$, then $ba = (-1)^{mn} ab$.

(3) A map of E_∞ -rings $f: A \rightarrow B$ is (faithfully) flat if:

(a) $\pi_0(f): \pi_0(A) \rightarrow \pi_0(B)$ is a (faithfully) flat map of ordinary commutative rings.

(b) For all $n \in \mathbb{Z}$, the induced map of $\pi_0(B)$ -modules

$$\pi_0(B) \otimes_{\pi_0(A)} \pi_n(A) \longrightarrow \pi_n(B)$$

is an isomorphism.

Descent vs hyperdescent

- > We'll start by explaining some subtle points about sheaf theory in this setting.
- > We'll first just work with topological spaces.

Def. Let X be a topological space and let $\mathcal{C}$ be an ∞ -category with limits. A presheaf $F: \text{Open}(X)^{\text{op}} \rightarrow \mathcal{C}$ is a sheaf if for every open $U \subset X$ and open cover $(U_i)_{i \in I}$ of U , the natural diagram

$$F(U) \longrightarrow \prod_{i_0 \in I} F(U_{i_0}) \rightrightarrows \prod_{i_0, i_1 \in I} F(U_{i_0} \cap U_{i_1}) \rightrightarrows \dots$$

is a limit diagram.

Lem. If $\mathcal{C} = \text{An}$ or Sp (more generally, a compactly generated ∞ -category), then the inclusion $\text{Sh}(X, \mathcal{C}) \hookrightarrow \text{Psh}(X, \mathcal{C})$ admits a left adjoint.

Def. Provided $\mathcal{C}$ has filtered colimits, for $x \in X$, the stalk functor $x^* \text{Sh}(X, \mathcal{C}) \rightarrow \mathcal{C}$ is the functor

$$F \mapsto \operatorname{colim}_{U \ni x} F(U)$$

Recall. If $\mathcal{C} = \text{Set}$, then a map of sheaves is an isomorphism if and only if it induces an isomorphism on stalks.

⚠ This is no longer true if $\mathcal{C} = \text{An}, \text{Mod}_{\mathbb{Z}}, \text{Sp}, \dots$

Exercise. Let $Q = \prod [0, 1]$ be the Hilbert cube. Show that Q has a basis of $\mathbb{N}$ opens abstractly homeomorphic to $[0, 1) \times Q$.

Ex. There is a sheaf

$$C_*^{\text{BM}}(-, \mathbb{Z}) : \text{Open}(Q)^{\text{op}} \rightarrow \text{Mod}_{\mathbb{Z}}$$

that assigns to each open U the complex of Borel-Moore chains on U .

> Since Q is compact, $C_*^{\text{BM}}(Q; \mathbb{Z}) \cong C_*(Q; \mathbb{Z}) \cong \mathbb{Z}[0]$.

> By the Künneth formula,

$$\begin{aligned} C_*^{BM}([0,1) \times Q, \mathbb{Z}) &\simeq C_*^{BM}([0,1), \mathbb{Z}) \bigotimes_{\mathbb{Z}} C_*^{BM}(Q, \mathbb{Z}) \\ &\simeq C_*^{BM}([0,1), \mathbb{Z}) \end{aligned}$$

- Borel-Moore homology can be computed as relative homology with respect to a compactification. Choosing $[0,1) \subset [0,1]$, we see that

$$C_*^{BM}([0,1), \mathbb{Z}) \simeq 0.$$

> Hence $C_*^{BM}(-, \mathbb{Z})$ vanishes on a basis, hence has 0 stalks.

- But $C_*^{BM}(Q, \mathbb{Z}) \simeq \mathbb{Z}[0]$, so the sheaf isn't zero!

Q. What's going on here?

Thm. Let X be a topological space and $F \in \text{Sh}(X, \mathcal{A}_n)$. TFAE:

- (1) For each map $G \rightarrow H$ in $\text{Sh}(X, \mathcal{A}_n)$ that induces an equivalence on stalks, the induced map

$$\text{Map}(H, F) \longrightarrow \text{Map}(G, F)$$

is an equivalence

- (2) F satisfies descent for hypercovers. This is slightly involved to formulate precisely, but basically means that for every open $U \subset X$, open cover $(U_i)_{i \in I}$ of U , collection of open covers $(V_j)_{j \in J_{i_0, i_1}}$ of $U_{i_0} \cap U_{i_1}$, etc., the diagram

$$F(U) \longrightarrow \prod_{i_0 \in I} F(U_{i_0}) \rightrightarrows \prod_{i_0, i_1 \in I} \prod_{j \in J_{i_0, i_1}} F(V_j) \rightrightarrows \dots$$

is a limit diagram.

- > In ring theory, the notion of a hypercover is a little easier to formulate.

Observation. Let $\mathcal{C}$ be an ∞ -category with limits, let $n \geq 0$, and write $i_n: \Delta_{\leq n}^{\text{op}} \hookrightarrow \Delta^{\text{op}}$ for the inclusion. The restriction functor

$$i_n^*: \text{Fun}(\Delta^{\text{op}}, \mathcal{C}) \longrightarrow \text{Fun}(\Delta_{\leq n}^{\text{op}}, \mathcal{C})$$

admits a fully faithful right adjoint $i_{n,*}$ given by right Kan extension along i_n :

$$(i_{n,*}(X))_k := \lim_{\substack{[m] \rightarrow [k] \\ m \leq n}} X_m$$

The composite $i_{n,*}i_n^*$ is called the n -coskeleton functor and denoted by

$$\text{cosk}_n: \text{Fun}(\Delta^{\text{op}}, \mathcal{C}) \longrightarrow \text{Fun}(\Delta^{\text{op}}, \mathcal{C}),$$

Ntn. Write $\text{Spec}: (\text{CAlg}^{\text{cn}})^{\text{op}} \hookrightarrow \text{Fun}(\text{CAlg}^{\text{cn}}, \text{An})$ for the Yoneda embedding, which carries colimits in CAlg^{cn} to limits.

> Lurie calls this Spét

Def Let $A \in \mathcal{CAlg}^{\text{cn}}$ and let $B_\bullet: \Delta \rightarrow (\mathcal{CAlg}^{\text{cn}})_A$ be a cosimplicial diagram under A . We say that $B_\bullet$ is an fpqc hypercovering of A if

(1) $A \rightarrow B_0$ is faithfully flat

(2) For each $n > 0$, the unit map

$$\text{Spec}(B_n) \rightarrow (\text{Cosk}_{n-1} \text{Spec}(B_\bullet))_n$$

is induced by a faithfully flat map.

We often denote an fpqc hypercovering by writing $A \rightarrow B_\bullet$.

Def. Let $\mathcal{C}$ be an ∞ -category with limits. A functor $F: \mathcal{CAlg}^{\text{cn}} \rightarrow \mathcal{C}$ is an:

(1) fpqc-sheaf if for every faithfully flat map of connective $\mathbb{E}_\infty$ -rings $A \rightarrow B$, the natural diagram

$$F(A) \rightarrow F(B) \rightrightarrows F(B \otimes_A B) \rightrightarrows \dots$$

is a limit diagram.

(2) fpqc-hypersheaf if for every fpqc-hypercover $A \rightarrow B_\bullet$, the natural map

$$F(A) \longrightarrow F(B_0) \rightrightarrows F(B_1) \Rrightarrow \dots$$

is an equivalence

Def We write $\text{Stk} \subset \text{Fun}(\text{CAlg}^{\text{cn}}, \text{An})$ for the full subcategory spanned by the fpqc-hypersheaves. We call objects of Stk stacks.

Quasicoherent sheaves

> We can define quasicoherent sheaves as before

Input The assignment $A \mapsto \text{Mod}_A = \text{Mod}_A(\text{Sp})$ extends to a functor

$$\text{Mod}_{(-)} : \text{CAlg}^{\text{cn}} \longrightarrow \text{CMon}(\text{Cat}_{\infty})$$

with functoriality given by relative tensor product.

Def The functor

$$\text{QCoh} : \text{Fun}(\text{CAlg}^{\text{cn}}, \text{An})^{\text{op}} \longrightarrow \text{CMon}(\text{Cat}_{\infty})$$

is the unique limit-preserving functor whose restriction along Spec is $\text{Mod}_{(-)}$. Explicitly,

$$\text{QCoh}(X) = \lim_{X \cdot \text{Spec}(A) \rightarrow X} \text{Mod}_A$$

> All remarks in the classical setting apply verbatim here.

Thm(Lurie). The functor

$$\mathrm{Mod}_{(-)}: \mathrm{Cat}^{\mathrm{cn}} \longrightarrow \mathrm{CMon}(\mathrm{Cat}_{\infty})$$

is an fpqc-hypersheaf. As a consequence, the functor

$$\mathrm{QCoh}: \mathrm{Stk}^q \longrightarrow \mathrm{CMon}(\mathrm{Cat}_{\infty})$$

carries colimit diagrams of stacks to limit diagrams of symmetric monoidal ∞ -categories.

Quotient stacks & equivariant sheaves

Def. Let X be a stack equipped with an action of a monoid M in $\mathbf{Stk}$. We write X/M for the quotient of X by M in $\mathbf{Stk}$, i.e., the colimit of the action diagram

$$\begin{array}{ccccc} \begin{array}{c} \xrightarrow{\quad} \\ \xleftarrow{\quad} \\ \xrightarrow{\quad} \\ \xleftarrow{\quad} \\ \xrightarrow{\quad} \end{array} & M \times M \times X & \begin{array}{c} \xrightarrow{\quad} \\ \xleftarrow{\quad} \\ \xrightarrow{\quad} \\ \xleftarrow{\quad} \\ \xrightarrow{\quad} \end{array} & M \times X & \begin{array}{c} \xrightarrow{a} \\ \xleftarrow{\quad} \\ \xrightarrow{\quad} \\ \xleftarrow{\quad} \\ \xrightarrow{\quad} \end{array} X \\ & & & & \text{pr}_2 \end{array}$$

The ∞ -category of M -equivariant quasicoherent sheaves on X is $\mathrm{QCoh}(X/M)$. Sometimes this is denoted $\mathrm{QCoh}_M(X)$.

Observe. By the previous result, $\mathrm{QCoh}(X/M)$ is the limit of the diagram

$$\mathrm{QCoh}(X) \begin{array}{c} \xrightarrow{a^*} \\ \xleftarrow{\quad} \\ \xrightarrow{\quad} \\ \xleftarrow{\quad} \\ \xrightarrow{\quad} \end{array} \mathrm{QCoh}(M \times X) \begin{array}{c} \xrightarrow{\quad} \\ \xleftarrow{\quad} \\ \xrightarrow{\quad} \\ \xleftarrow{\quad} \\ \xrightarrow{\quad} \end{array} \mathrm{QCoh}(M \times M \times X) \begin{array}{c} \xrightarrow{\quad} \\ \xleftarrow{\quad} \\ \xrightarrow{\quad} \\ \xleftarrow{\quad} \\ \xrightarrow{\quad} \end{array} \dots$$

pr_2^*

So to first approximation, an M -equivariant quasicoherent sheaf is a quasicoherent sheaf $F \in \mathrm{QCoh}(X)$ together

With an equivalence $a^*(F) \simeq \text{pr}_2^*(X)$, subject to all possible higher-order coherences.

Upshot. Given a fan (N, Σ) , we can do the same construction as before to define the toric stack X_Σ over S . We define X_Σ as the colimit of

$$\begin{array}{ccc}
 \Sigma^q & \longrightarrow & \text{CMon}(\text{Set}) \hookrightarrow \text{CMon}(\text{An}) \xrightarrow{S \leftrightarrow} \text{bCAlg}(\text{Sp}^{\text{cn}}) \\
 \sigma_1 & \longrightarrow & \sigma^\vee \cap M
 \end{array}
 \begin{array}{c}
 \\
 \\
 \downarrow \text{Spec} \\
 \text{CMon}(\text{Stk}) \\
 \downarrow \text{forget} \\
 \text{Stk}
 \end{array}$$

> The torus T_N is the commutative group stack

$$T_N := \text{Spec}(S\langle M \rangle).$$

> X_Σ has a natural T -action, and we'll work with X_Σ/T .